

POLITECNICO DI TORINO

MSc Technical Activity

What Corrosion Photographs Can Support

Surface assessment, terminal capacity,
and the limits of inference in ferrocement

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Thesis-style manuscript

Version 3 | 16 September 2026

A synthesis and re-verification of saved modelling experiments on a predecessor experimental dataset. Prepared for scientific review; programme-specific submission metadata and approval are not asserted.

Abstract

Photographs can reveal visible corrosion without establishing either internal damage or remaining service life. This technical activity examines that distinction in ferrocement using a predecessor dataset of 48 specimens from two experimental campaigns. The current aligned basis contains 791 readable photographic observations, but wire-area loss and ultimate load are each measured only once per specimen, at terminal testing. Campaign identity is aligned with mesh configuration, chloride concentration, and terminal exposure duration.

Saved experiments are synthesized by scientific question and independently checked through their tables, predictions, and split manifests. Classical colour features, frozen residual-network embeddings, and interpretable image descriptors support visible-corrosion estimation under specimen-grouped evaluation. Some strong results are label-adjacent: total-rust coverage and a rust-area feature have Spearman correlation 0.9996. This limits their interpretation as independent predictive discovery.

The focused terminal-load experiment compares image-only, metadata-only, and combined feature families. Scoring uses held-out terminal photographs, not a fixed early observation horizon. A metadata-only Ridge model has mean terminal-test MAE of 0.173 kN, with fold standard deviation 0.046 kN, across ten folds covering 48 specimens. Adding image descriptors does not establish a reliable pooled improvement. A post-onset sensitivity analysis has a limited gain in MAE for a combined configuration, without uniformly better ranking. Within each mesh family, every selected feature-set model has negative mean test R^2 . Campaign exclusion is a severe extrapolation stress test, not an ordinary estimate of deployment performance.

Model-derived structural trajectories support only exploratory threshold-status analysis. The refined configuration has zero future threshold crossings, while crossings at baseline and during observation remain present. No lifetime-event supervision exists. The separate four-class severity branch has completed data preparation but no trained model. The principal contribution is therefore a bounded account of what the observations and evaluation design support, together with a concrete path toward stronger validation.

Executive Summary

Scientific conclusion. Surface assessment is the most defensible image-based task in this dataset. The available experiments do not establish transferable structural-condition inference or remaining-life prediction. This is a result about the present observations, representations, and evaluation procedures, not an impossibility claim about imaging.

Why the distinction matters. A photographed surface, a destructively measured wire cross-section, and a terminal bending load are different quantities. Repeated images provide rich visual supervision; repeating a specimen’s terminal target alongside those images does not create repeated structural measurements. A model can exploit real design information and campaign-specific associations simultaneously.

What supports the conclusion. The most mature load analysis uses a complete feature-family comparison. Its pooled metadata-only result is competitive with all image-enhanced alternatives. The loss of performance within a mesh family and across campaigns limits the interpretation of the pooled score. Surface-treatment transfer is phase-dependent: one interpretable representation deteriorates under treatment exclusion, while a later representation does not show the same pattern.

What remains useful. The work provides reproducible aggregation of saved results, specimen-disjoint evaluation manifests, a clear separation between observed and derived quantities, and a prepared classification dataset. Negative results identify precisely which claims require new evidence.

Next research decisions. Complete the four-class classifier as a separate, controlled experiment; repair current configuration and source defects before retraining; evaluate early-image prediction at predeclared time horizons; and collect crossed experimental designs and independent structural or lifetime outcomes for stronger structural and prognostic claims. The report rebuild itself performs no model training.

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Chapter 1

Introduction

This study asks how far repeated corrosion photographs can support an inference about ferrocement condition. Its central concern is the boundary between estimating an observable surface quantity and inferring a structural endpoint that has much less supervision.

1.1 Motivation and problem

Ferrocement combines a cementitious matrix with distributed steel-mesh reinforcement. For the laboratory specimens considered here, external rust evidence can be recorded repeatedly while structural testing is destructive. This asymmetry makes visual monitoring attractive but complicates the interpretation of an image-based predictor. An accurate estimate of visible corrosion is not automatically an accurate estimate of lost wire area, residual capacity, or future service life.

The physical specimens, exposure campaigns, and terminal experiments originate in predecessor work [? ?]. The present activity concerns subsequent data preparation, image representations, supervised models, evaluation, and interpretation of the saved computational results. It does not claim authorship of the original experimental campaign or a fresh reproduction of its image-processing protocol.

A recurring scientific question is whether a model learns a transferable relationship or a feature of the experimental mixture. If each campaign uses its own mesh configuration and exposure conditions, a strong pooled capacity score can partly reflect the separation between campaigns. Testing on new specimens from that mixture and testing on an unseen campaign therefore ask materially different questions.

1.2 Research questions

The report organizes the evidence around five questions:

- RQ1.** How consistently can visible surface corrosion be estimated across distinct image representations, and how does label adjacency limit that interpretation?
- RQ2.** How well do the available inputs predict the two terminal structural endpoints, and how stable are the results under stricter evaluation?
- RQ3.** Do image-derived corrosion descriptors add predictive value beyond specimen and exposure metadata for terminal ultimate load?
- RQ4.** How do grouped specimen evaluation, treatment exclusion, campaign exclusion, and mesh stratification change the defensible claim?
- RQ5.** What does a model-derived degradation and threshold-crossing pipeline establish when measured structural trajectories and lifetime outcomes are absent?

The prepared four-class classification branch is reported as a development status, not as a sixth answered performance question.

1.3 Contributions and scope

The first contribution is a synthesis that separates target observability from model sophistication. The same image collection supports a well-supervised surface task and a sparsely supervised structural task; these should not be assessed with the same evidential expectations.

The second contribution is a focused interpretation of the terminal-load ablation. Image-only, metadata-only, and combined models are compared under the same specimen partitions, with the scope of the prediction target made explicit. The experiment assesses terminal load from terminal held-out observations, even though training uses the broader image history with repeated terminal labels.

The third contribution is a verification and reporting framework for the saved evidence. It checks specimen disjointness, reproduces key metrics from saved predictions, resolves conflicting model-selection summaries, and documents figure provenance. This supports reproduction of the reported aggregation; it does not establish that the historically trained models can currently be regenerated without source repairs.

The conclusions are restricted to the available specimens, representations, and evaluation designs. They do not identify the independent causal effects of mesh, chloride, or duration. They do not validate a structural safety decision, a remaining-life estimator, or a four-class classifier. They also do not establish that surface images contain no structural information: the distinction is between a physical relationship that may exist and a stable predictive relationship demonstrated by these experiments.

1.4 Organization

Chapter 2 positions the study in imaging and evaluation research. Chapters 3 and 4 define observations and methods. Chapters 5–7 answer the research questions using saved evidence. The discussion and limitations then connect those observations to concrete research decisions. Reproduction details are confined to Appendix A so the scientific narrative remains readable.

Chapter 2

Background and Related Work

The relevant literature concerns both the physical meaning of an image-derived corrosion measure and the evaluation of learned relationships. This chapter uses a focused selection of verified sources to establish those distinctions; it is not a systematic literature review.

2.1 Predecessor experimental and imaging work

Hossain’s thesis describes the experimental ferrocement specimens, exposure procedures, photographic record, mechanical testing, and microscopic investigation [?]. In that work, four-point bending is performed after ageing; the microscopic assessment examines longitudinal wires in the layer near the cracked or exposed surface. These measurements provide the physical context for the terminal endpoints used here. A selected cross-sectional area-loss measurement does not constitute an observed internal-damage trajectory over the exposure period.

The related conference paper develops an image-analysis and classification workflow for ferrocement surface corrosion [?]. Its RGB-based analysis and three-category classification belong to that external predecessor study. The present repository contains different computational generations and a later four-class preparation branch. Their class schemes, data handling, and trained-model status must remain separate. The predecessor’s classifier results cannot be used as results of the current four-class branch.

The contribution of the present activity is therefore a secondary modelling and evaluation study on the inherited observations. Reusing this experimental basis allows representation and evaluation comparisons, but does not provide an independent replication of the physical experiment.

2.2 Image-based corrosion and structural monitoring

Atha and Jahanshahi evaluate convolutional-network approaches to corrosion detection [?]. Their work motivates visual feature learning for detecting corrosion evidence. It does not establish that a detector’s score estimates internal metal loss or terminal capacity in ferrocement. Those are different targets requiring their own ground truth and validation.

Spencer, Hoskere, and Narazaki distinguish inspection applications from monitoring applications in their review of computer vision for civil infrastructure [?]. That distinction is useful here: identifying visible damage and measuring structural response need different observation models. Repeated photographs can support an inspection time series, but the availability of a time series alone does not make its downstream structural interpretation longitudinally validated.

Frozen residual-network embeddings provide one intermediate representation between colour rules and fully task-specific deep learning. Residual networks were developed to facilitate the optimization of deep visual models [?]; in this activity, an existing network supplies fixed image features and a separately trained regression head predicts the visible label. The saved experiment is evidence about that particular transfer-learning arrangement, not about the universal superiority of deep or handcrafted representations.

2.3 Evaluation must match the prediction question

Repeated observations create a dependence structure. Roberts et al. discuss cross-validation for structured data and show why the blocking strategy must reflect the intended prediction problem [?]. Their methodological discussion also matters for interpretation: withholding a group can reduce overlap in the predictor space. Here, specimen exclusion prevents a specimen’s image history from appearing in both fitting and evaluation, while campaign exclusion additionally removes a whole design/exposure combination.

Kapoor and Narayanan connect leakage risks to the validity of scientific claims made from machine-learning performance [?]. In this study the essential information-flow questions are concrete: whether a specimen crosses partitions, whether terminal targets enter the feature matrix, whether preprocessing uses held-out data, and whether model-generated structural estimates are subsequently mistaken for observations. A clean specimen split addresses one of these questions, not all of them.

Model selection creates an additional boundary. Cawley and Talbot show that optimizing a finite-sample evaluation criterion can introduce selection bias [?]. Accordingly, a selected winner’s reported fold score should not be presented as an unbiased estimate of the full model-selection procedure. This is especially pertinent when image-family differences are small and the number of independent structural observations is limited.

2.4 Position of this study

The evidence reviewed above motivates a specific contribution: testing whether visible-corrosion representations support incremental structural prediction under realistic information and grouping constraints. The specific question is incremental terminal-capacity estimation after accounting for metadata, rather than visual corrosion detection alone. The study supplies a bounded empirical case, not a new general-purpose model; the focused literature search does not establish an exhaustive novelty claim. Its value lies in explaining how apparently encouraging pooled scores change when the target, observation unit, and evaluation population are specified. The next chapter makes that observation structure explicit.

Chapter 3

Dataset and Experimental Context

The meaning of every result depends on what was measured and when. This chapter establishes the specimen, campaign, and supervision structure before any model score is considered.

3.1 Experimental units and alignment

The current aligned dataset contains 791 photographic observations from 48 physical specimens. The raw archive contains 792 PNG files; one is unreadable. Independent archive verification reproduced the saved readability audit. Specimens have 15–18 retained observations, with a median of 16, across an overall range of weeks 0–36. These observations span 25 distinct week values, reflecting the combined schedules rather than a common uniformly sampled sequence.

The experimental unit for structural inference is the specimen. Images of one specimen share design, exposure, surface characteristics, and the eventual destructive test. Counting each image as an independent structural sample would overstate the information available.

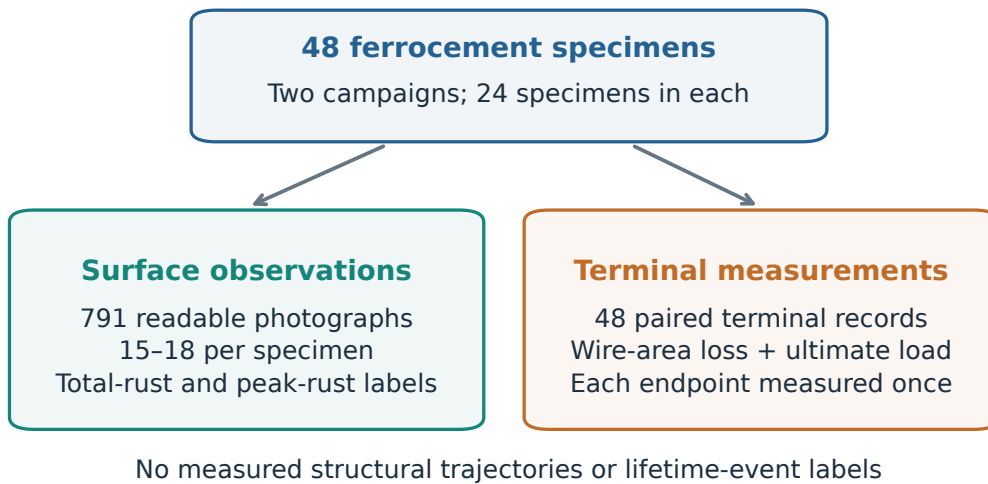


Figure 3.1: Dataset and supervision structure on the current aligned basis. There are 791 readable photographs across 48 specimens. Visible total-rust and peak-rust labels accompany repeated observations; wire-area loss and ultimate load are each measured once per specimen at terminal testing. Arrows summarize acquisition and label availability, not a causal model. Counts are reproduced from the saved aligned table.

3.2 Campaign design and confounding

There are two campaigns with 24 specimens each. Table 3.1 records the observed combinations of the main design and exposure variables. Treatment protocols vary within campaigns, so these are two campaign-level combinations of selected factors, not two identical sets of specimens. No observation combines the first campaign’s mesh count with the second campaign’s chloride and duration. A predictor using any of these factors can therefore identify the same broad division of the dataset. Their separate causal contributions cannot be recovered from their pooled associations in this design.

Table 3.1: Observed campaign-level design combinations for all 48 specimens. Mesh count, chloride concentration, and terminal exposure do not vary independently across campaigns. Exposure days here are derived from the recorded terminal week; they are not a claim about failure time.

Campaign	Specimens	Mesh layers	NaCl (%)	Terminal week (days)
1	24	7	3.5	28 (196)
2	24	4	5.0	36 (252)

3.3 Targets and units

Visible total rust is the recorded fraction of the inspected surface expressed in percent. **Peak rust** describes the most affected local region, also expressed in percent. They are distinct targets: a feature aligned with total coverage does not automatically establish the same numerical identity with the peak label.

Wire-area loss is a terminal cross-sectional measurement reported as a fraction in the current aligned table. The workbook’s percentage-like column name is misleading: multiplying the stored fraction by 100 produces percent. Comparisons with an earlier representation that uses percentage units require this conversion. MAE of a fraction and MAE in percentage points must not be placed on the same numerical scale without it.

Ultimate load is the terminal maximum load from bending, expressed in kilonewtons. The predecessor thesis describes four-point bending after ageing and the microscopic examination of a selected longitudinal wire layer near the crack surface [?, secs. 4.2.2.3, 4.2.3, and 5.2.1]. Both wire-area loss and load are directly measured terminal endpoints. Neither provides a repeated measurement of structural state through exposure.

3.4 Supervision asymmetry and target reuse

All retained image observations have the visible targets, whereas only 48 rows carry each measured structural endpoint. These structural rows coincide with the terminal observation for each specimen. There is no observed service-life endpoint, longitudinal structural trajectory, or time-to-failure supervision.

The later capacity workflow joins the terminal load back onto earlier images of the same specimen for training. This is a target assignment for a supervised experiment, not an additional physical measurement. The assigned label always means eventual terminal load; it must not be read as the specimen’s measured capacity at the image’s earlier time.

3.5 Dataset versions and classification balance

Earlier modelling generations retained a 792-row basis and used fallback or imputed image features for the unreadable image. The current basis excludes that observation. Historical categorical schemas also differ: an earlier interpretable phase uses five categories, while the current total-rust schema has four. Consequently, historical classification metrics are not transferable to the current branch.

The current total-rust class counts are 658, 99, 20, and 14 for classes 1–4. This imbalance remains a design consideration even after uniform augmentation: generating the same number of transformed copies per original does not increase the number of independent specimens or equalize class frequencies. Chapter 10 reports the actual retained split inventory.

The following chapter defines specimen-disjoint evaluation and the population tested by each regime.

Chapter 4

Methodology and Evaluation

This chapter defines the prediction tasks, information flow, and comparisons used to interpret the saved experiments. No models were retrained for the report; numerical checks operate on saved outputs.

4.1 Notation and prediction targets

Let i identify a physical specimen, t an observation time, I_{it} its image, and m_{it} the available design and exposure metadata. Let v_{it} denote a visible-corrosion label, a_i terminal wire-area loss, and u_i terminal ultimate load. A representation $\phi(I_{it})$ converts an image into model inputs.

Visible-corrosion regression estimates

$$\hat{v}_{it} = f(\phi(I_{it}), m_{it}), \quad (4.1)$$

with image-only variants omitting metadata. Terminal-capacity regression instead targets

$$\hat{u}_{it} = g(\phi(I_{it}), m_{it}), \quad y_{it} = u_i. \quad (4.2)$$

The repeated label in Eq. (4.2) is terminal load at every training time. The main capacity score is calculated at the measured terminal image of each held-out specimen. It does not measure performance at a fixed early observation horizon.

Table 4.1: Information availability in the focused capacity analysis. The primary evaluation is retrospective at the terminal image. A future early-image experiment must verify that every predictor is available before its chosen cutoff.

Quantity	Role and timing boundary
Image descriptors	Derived from the image at its observation time.
Design/treatment metadata	Recorded specimen context; availability is an assumption to check for a future deployment.
Week and ageing days	Exposure context at image acquisition.
Failure-surface cover	Included metadata; pre-test availability is not established by the saved feature definition.
Terminal ultimate load	Training target, repeated over image rows; evaluation uses terminal observations. Excluded from predictors.
Wire-area loss / visible labels	Excluded predictors in the focused capacity experiment; use in other phases is explicitly distinguished.

4.2 Three evaluation regimes

All train/test partitions must keep a specimen’s observations together. The audited saved manifests satisfy this disjointness requirement. Figure 4.1 illustrates the unit being withheld.

Grouped specimen holdout withholds specimens sampled from the observed experimental mixture. Unstratified grouped splitting does not guarantee matching proportions of campaigns or treatments. The later capacity analysis explicitly stratifies specimen folds by mesh in the pooled case.

Leave-one-treatment-out (LOTO) withholds a treatment group. Its interpretation depends on the treatment definition: the earlier interpretable phase uses seven coarse groups, while the refined surface experiment uses nine more specific groups. Their results therefore cannot be treated as identical tests of a single invariant property.

Leave-one-campaign-out (LOCO) fits on one campaign and evaluates on the other. With two campaigns, there are two directions of transfer. Because campaign, mesh, chloride, and terminal duration coincide, this is a severe extrapolation stress test. The term does not imply a representative sample of all future sites or campaigns.

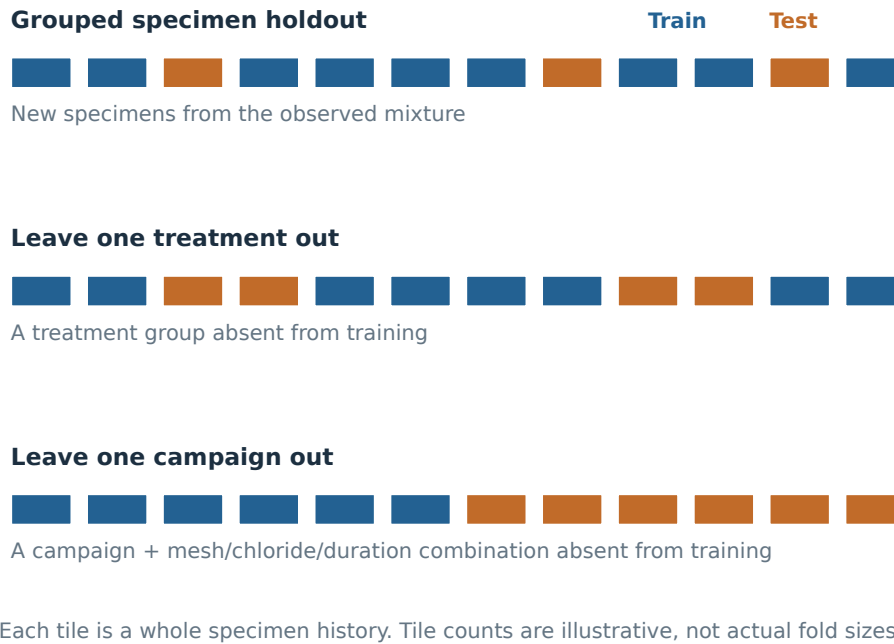


Figure 4.1: Evaluation regimes with specimen histories kept intact. Grouped holdout samples specimens from the observed mixture; LOTO withholds a treatment definition; LOCO withholds a campaign and its aligned design/exposure combination. Tile counts are illustrative and do not represent actual fold sizes. The actual manifests, including mesh-stratified and post-onset analyses, were checked for specimen overlap.

4.3 Image representations and model families

The earliest exploratory scripts demonstrate a workflow but supply no held-out benchmark used here. The classical benchmark uses colour statistics and threshold-derived mask descriptors with regression models. The deep representation uses a frozen ResNet-18 encoder and a learned regression head; its two configurations are image-only and image plus metadata. It has no metadata-only baseline.

The interpretable multi-family phase combines spatial rust summaries, colour, texture, and morphology. Its structural models also include metadata and recorded visible labels; they are mixed-input models, not image-only estimators. The later robustness analysis evaluates cleaned feature sets and selects structural configurations using a weighted rank criterion across regimes.

The focused load experiment compares metadata, RGB, HSV, RGB+HSV, and each corresponding metadata combination. Its regressors include Ridge, RandomForest, GradientBoosting, XGBoost, and CatBoost. Categorical metadata are imputed and encoded within training folds; Ridge inputs are standardized using the training data. Image labels, measured structural targets, identifiers, and future-derived summary fields are excluded from the input feature list of this focused analysis. Detailed feature names and source locations appear in Appendix A.

4.4 Terminal-load comparison design

The pooled load experiment uses two repetitions of five specimen folds, covering all 48 specimens in each repetition. Each held-out fold contains nine or ten terminal specimens. Training uses all retained image rows belonging to training specimens, while metrics use the terminal observations of test specimens. The resulting image-row objective weights specimens in proportion to the number of retained photographs unless explicitly corrected; the saved primary workflow does not apply such correction.

Models are selected separately within feature families using the saved evaluation scores. This comparison asks whether image-enhanced configurations consistently improve on a metadata anchor under a matched partition design. Since the same folds help select and report the winners, the results remain exploratory estimates for selected configurations, not a nested evaluation of the complete selection procedure [?].

Two sensitivity analyses constrain the pooled interpretation. The first repeats the comparison within each mesh family, leaving 24 independent structural specimens per family and only four or five terminal test specimens per fold. The second retains observations at and after a specimen's first recorded total-rust value of at least 0.5%, leaving 43 specimens with eligible terminal observations and eight or nine terminal test specimens per fold. The threshold is a pragmatic sensitivity cutoff. The code retains all later observations once onset occurs; it does not require each later value to remain above the cutoff.

A separate specimen-summary analysis first predicts load using metadata and then attempts to correct residuals with corrosion summaries within mesh groups. Its training-stage cross-fitting concerns that refinement experiment. It must not be confused with the in-sample structural estimates that feed the degradation pipeline.

4.5 Metrics, aggregation, and uncertainty

For a test set of n observations with errors $e_j = \hat{y}_j - y_j$,

$$\text{MAE} = \frac{1}{n} \sum_j |e_j|, \quad \text{RMSE} = \sqrt{\frac{1}{n} \sum_j e_j^2}. \quad (4.3)$$

MAE measures typical absolute error in the target's units; RMSE places more weight on larger errors. The coefficient of determination is

$$R^2 = 1 - \frac{\sum_j e_j^2}{\sum_j (y_j - \bar{y}_{\text{test}})^2}. \quad (4.4)$$

Negative test R^2 means squared error exceeds the test-set-mean reference for that fold. That reference is a score definition, not a fitted deployable baseline. In small, narrow-range subgroups its denominator can be small, making fold R^2 unstable. Spearman correlation ρ evaluates rank agreement and supplies complementary information, but is also variable in small folds.

The main load tables report arithmetic means across ten folds and, where shown, sample standard deviations across those folds. Repetitions share specimens and training sets; their standard deviation is not a confidence interval on future error. An average of fold scores also need not equal a score calculated after averaging a specimen's repeated predictions. Neither a small difference nor overlapping descriptive error bars establishes statistical equivalence.

4.6 Fixed models versus reselected winners

Two legitimate summaries answer different questions. A per-regime winner describes the best selected configuration under that regime. A fixed-model robustness comparison follows the same grouped-holdout winner into each stricter regime. Only the latter supports a ratio of regime MAE to that model's own grouped MAE. Cross-generation comparisons in this report use such within-generation ratios or qualitative patterns, rather than a common raw-error ranking.

These definitions keep observations, model selection, and transfer questions separate. The next two chapters apply them first to the visible targets and then to terminal structural outcomes.

Chapter 5

Results: Visible Surface Corrosion

This chapter answers RQ1 and the surface component of RQ4. The evidence supports learnable visible corrosion, while also showing why a highly accurate visual benchmark must be checked for label adjacency.

5.1 Signal across distinct representations

The classical threshold-feature benchmark, frozen image embedding, and interpretable multi-family representation all give positive grouped-test R^2 for peak-rust estimation. Table 5.1 reports selected saved results with their own evaluation contexts. The rows are not an ordered competition: they differ in representation, fitting procedure, and selection. The embedding experiment is useful because its

Table 5.1: Historical peak-rust estimation on the 792-row, 48-specimen basis. MAE and RMSE are in percentage points. The classical row is its selected current-corrosion model under grouped holdout, with 159 test observations from ten specimens reconstructed from the saved IDs and split definition; the embedding rows use the same held-out 159 observations from ten specimens; the interpretable row is its grouped-holdout winner on 159 observations. These protocols are not identical, so the table establishes signal across representations rather than a causal ranking of representations.

Representation	Model	MAE	RMSE	R^2
Classical + metadata	RandomForest	2.379	4.254	0.900
Frozen embedding	Image-only MLP	5.626	9.264	0.524
Embedding + metadata	Multimodal MLP	5.748	9.865	0.460
Interpretable mixed features	MLP regressor	4.209	6.234	0.784

image-only configuration demonstrates signal without specimen metadata in the regression input. Its comparison with the multimodal configuration tests adding metadata to images. It cannot answer whether images improve a metadata model: that experiment lacks a metadata-only anchor.

These representations are distinct computational approaches applied to overlapping specimens. Their agreement is corroboration within the same experimental resource, not three independent physical replications. Nor does a positive score establish field robustness under different cameras, substrates, illumination, or exposure mechanisms.

5.2 Near reconstruction of a surface label

In the refined feature audit, total surface-rust percentage and the independently named rust-area image feature have Spearman correlation 0.9996 across all 791 aligned observations. Their mean absolute difference is approximately 0.00023 percentage points, with a maximum of approximately 0.00050 percentage points. Figure 5.1 shows both the agreement and the small residuals.

The generation boundary matters. In the earlier interpretable feature table, the total-rust label and rust-area descriptor have Pearson correlation 0.257; the corresponding peak-rust and peak-location correlations are 0.183 and 0.248. These values are independently recomputed on 792 matched historical rows and agree with the earlier feature-alignment figure. They concern different descriptors from the

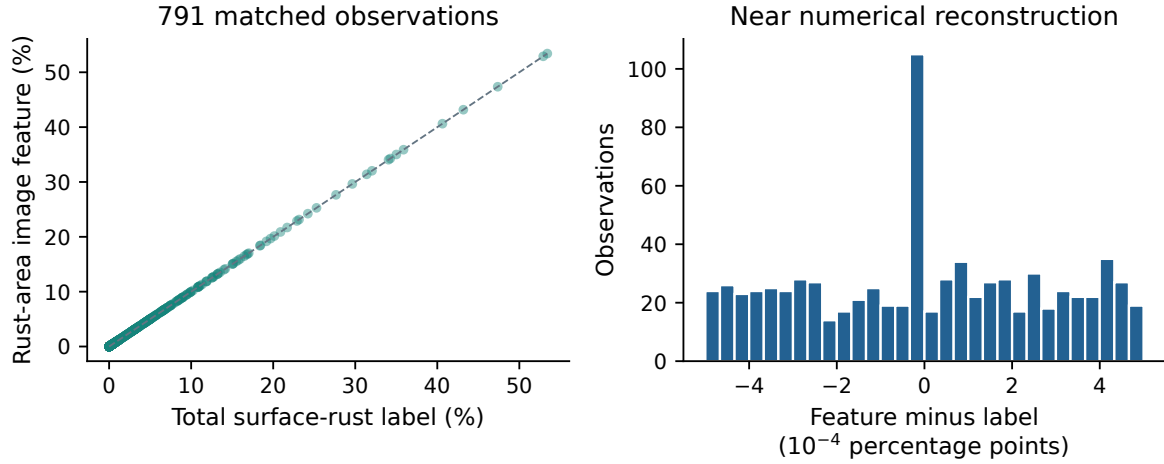


Figure 5.1: Numerical alignment of the total surface-rust label and the rust-area feature on all 791 matched observations. The left panel compares the two percentages with an identity line; the right panel shows feature-minus-label residuals in percentage points. Spearman correlation is 0.9996. This is evidence of near label reconstruction; it does not independently verify that identical code or thresholds created the original label.

refined near-identity comparison above. The label-adjacency conclusion therefore applies to the refined representation, not automatically to every image-feature generation.

The refined near-identity result is useful as a measurement-consistency check. It is weak evidence for a broad claim that a model discovered a new relation between appearance and corrosion state. Learning a nearly equivalent feature-to-label mapping is different from recovering an independent physical target. The classical peak-rust benchmark has a related concern because a colour-mask feature dominates its importance ranking, but that is a different target and the available evidence does not establish the same numerical identity.

Peak rust remains the more informative surface benchmark in the refined analysis, although its strongest spatial rust descriptors are themselves closely related to the quantity being measured. This is not automatically an invalid task: automating a visual measurement can be useful. The claim should be measurement estimation within the observed label system, not inference of hidden structural damage.

5.3 Regime sensitivity is phase-dependent

For the earlier interpretable representation, the per-regime selected total-rust model has mean $R^2 = 0.582$ under campaign exclusion and -0.677 under treatment exclusion. Corresponding peak-rust values are 0.595 and -0.077 . These are means over two campaign folds and seven treatment folds, respectively, on the 792-row basis. Individual treatment test sets are small and uneven; averaging their R^2 values can amplify a poor low-variance fold.

The refined surface experiment does not reproduce this treatment-level deterioration. Tracking its grouped winners, total-rust GradientBoosting has LOTO/grouped MAE ratio 0.549 , while peak-rust RandomForest has ratio 0.758 . Their mean treatment-fold R^2 values are 0.995 and 0.957 across nine folds. The corresponding LOTO Spearman values in the saved robustness table are 0.989 and 0.972 . These results apply to the refined 791-row basis and its more specific treatment grouping.

Figure 5.2 places the surface and structural patterns side by side using fixed models. Its ratios should not be confused with the reselected-model R^2 values above. For example, a positive campaign-out R^2 for the best available peak-rust model does not imply that its original grouped winner keeps the same absolute error.

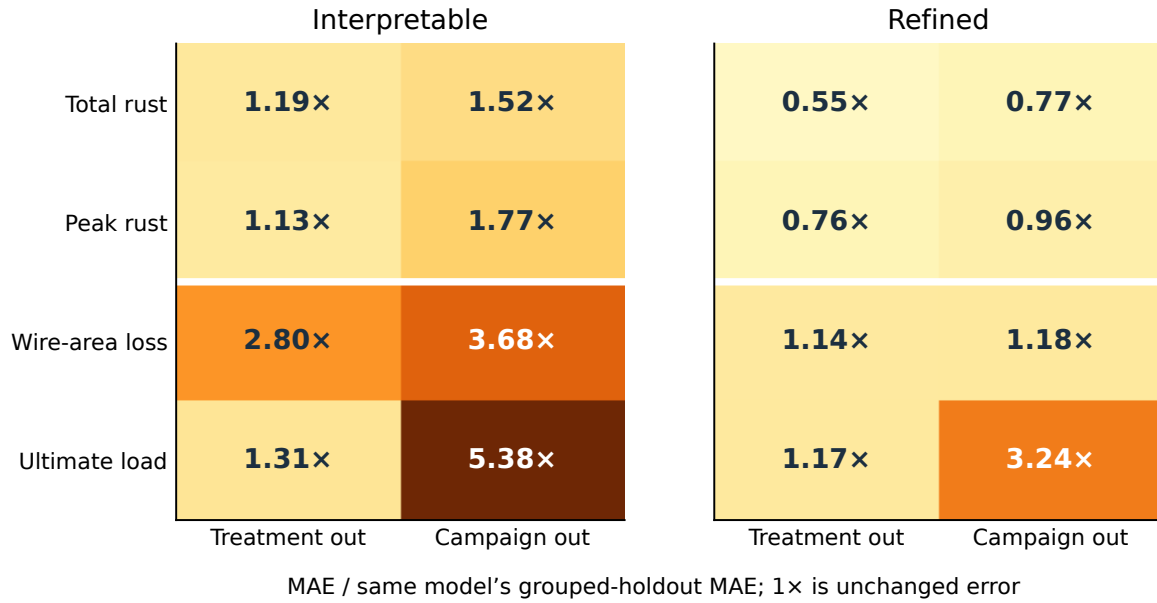


Figure 5.2: Evaluation-regime sensitivity expressed as MAE divided by each model's own grouped-holdout MAE. Values above one indicate larger error. For the interpretable representation, fixed grouped winners are RandomForest for total rust, wire-area loss, and ultimate load, and MLP for peak rust. For the refined representation they are GradientBoosting for total rust and RandomForest for the other targets; the wire-loss row is the diagnostic winner, not the final robustness-selected CatBoost. Earlier grouping uses one holdout, seven treatment folds, and two campaign folds; refined grouping uses five holdouts, nine treatment folds, and two campaign folds. Structural results have 48 independent terminal specimens. The generations are normalized separately.

5.4 Answer to the surface question

Visible corrosion is consistently learnable within the saved experiments, but the strongest total-coverage result is almost reconstructive. Campaign-out surface results remain informative in both later representations; treatment-out results differ between them. The available evidence does not isolate whether representation, label/schema changes, or treatment grouping explains that disagreement. The defensible conclusion is phase-dependent treatment robustness, followed by a separate assessment of structural targets in the next chapter.

Chapter 6

Results: Terminal Structural Condition

This chapter answers RQ2–RQ4 for the measured structural endpoints. It first distinguishes robust model selection from diagnostic winners, then concentrates on the strongest controlled question: the incremental value of image features for terminal load.

6.1 Two measured endpoints, weaker transfer

In the earlier interpretable phase, grouped-holdout RandomForest gives wire-area-loss MAE 3.899 percentage points and $R^2 = 0.310$, while ultimate-load MAE is 0.121 kN and $R^2 = 0.385$. These scores use ten test specimens from 48, with 38 training specimens. They show some within-mixture predictability, but do not establish stable transfer.

The later robustness-weighted selection chooses metadata-only models for both endpoints (Table 6.1). For wire-area loss the final selection is CatBoost, with grouped MAE 0.124 as a fraction. A separate diagnostic table follows a RandomForest with grouped MAE 0.105. Those rows must not be merged: the lower diagnostic number is not the final CatBoost model’s score. The final load model’s campaign-out

Table 6.1: Final robustness-selected structural configurations on the 791-image basis with 48 terminal specimens. Both use metadata only. Entries are mean MAE across five grouped holdouts, nine treatment folds, or two campaign folds; wire-loss error is a fraction, and load error is in kN. Selection weighs evidence across regimes, so these are not necessarily the smallest isolated grouped errors.

Target	Model	Grouped	LOTO	LOCO
Wire-area loss (fraction)	CatBoost	0.124	0.116	0.121
Ultimate load (kN)	RandomForest	0.174	0.203	0.564

MAE is 3.236 times its own grouped MAE, and its mean rank correlation falls from 0.778 to 0.353. The diagnostic wire-loss RandomForest has campaign-out Spearman -0.089 . This combination of modest within-mixture error and poor transfer motivates the focused load study, rather than a direct image-to-life claim.

6.2 Do images improve the metadata baseline?

Table 6.2 gives all seven selected feature families in the pooled terminal-load experiment. Metrics use terminal held-out observations, with 48 specimens covered in each of two five-fold repetitions. The metadata-only Ridge configuration has mean MAE 0.173 kN, fold SD 0.046 kN, mean RMSE 0.214 kN, and mean Spearman 0.758.

Table 6.2: Pooled terminal-load comparison for 48 specimens across ten grouped folds. MAE and its sample SD are in kN; R^2 and Spearman ρ are arithmetic means of fold metrics. Each feature family shows its selected model, chosen using the same saved folds. The table is an exploratory comparison, not a nested estimate of the model-selection procedure or a significance test.

Features	Model	MAE	SD	R^2	ρ
metadata-only	Ridge	0.173	0.046	0.645	0.758
metadata+HSV	Ridge	0.173	0.039	0.635	0.771
metadata+RGB+HSV	CatBoost	0.183	0.057	0.607	0.753
metadata+RGB	Ridge	0.184	0.047	0.584	0.748
HSV-only	CatBoost	0.210	0.054	0.481	0.705
RGB+HSV	CatBoost	0.234	0.056	0.411	0.668
RGB-only	CatBoost	0.253	0.059	0.269	0.501

Adding HSV descriptors to metadata changes mean MAE only from 0.172711 to 0.173003 kN while increasing mean rank correlation from 0.758 to 0.771. The MAE change is about 0.000293 kN. It is not evidence of an improvement, and the available comparison does not establish equivalence either. Image-only feature families have larger pooled MAE than the metadata anchor. The combined RGB+HSV configuration with metadata is also higher in MAE than the anchor.

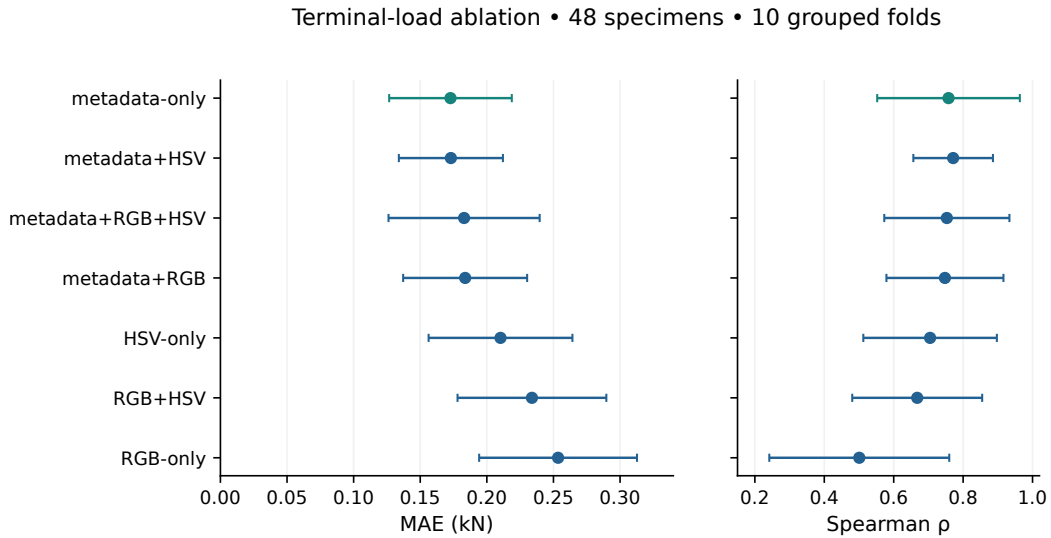


Figure 6.1: Pooled terminal-load results for selected feature-family models on 48 independent specimens across ten grouped folds. Points are arithmetic means and horizontal bars are sample standard deviations across folds, not confidence intervals. Terminal test sets contain nine or ten specimens. All training photographs share their specimen's terminal target; scoring is at terminal observations. Models are identified in Table 6.2. The figure summarizes the same saved comparison, not an additional experiment.

The observed pattern supports a bounded conclusion: the tested image descriptors do not provide a reliable pooled increment over metadata for terminal load in this experiment. It does not show that images are irrelevant to structural behaviour. It also does not show that metadata effects are causal or transferable outside the experimental mixture. Section 6.3 examines what the same baseline gets wrong and what its saved fit can reveal.

6.2.1 Paired specimen errors with the model family fixed

The near-tie can be examined without retraining by pairing metadata-only Ridge and metadata+HSV Ridge predictions for the same specimen in the same fold. For each specimen, absolute terminal errors

are averaged across its two held-out evaluations. The difference is defined as image-enhanced error minus metadata-only error, so a negative value favours adding HSV features.

Adding HSV improves 19 specimens and worsens 29. The mean paired difference is approximately $+0.0002$ kN and the median is $+0.0094$ kN. Figure 6.2 shows that the near-zero mean combines improvements and worsenings. These quantities weight each specimen equally, whereas Table 6.2 weights each fold equally; the small difference between the two mean-difference calculations is expected. This is a descriptive analysis of selected saved models, not a significance test or an equivalence analysis.

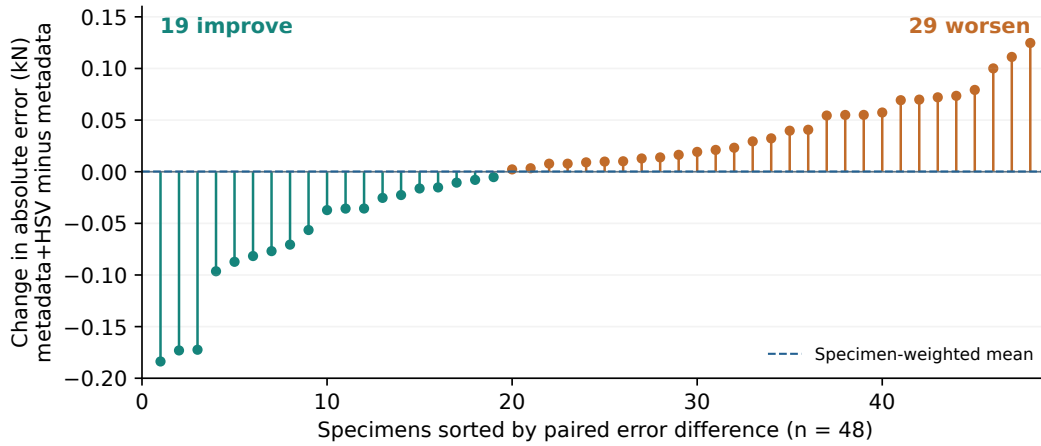


Figure 6.2: Paired effect of adding HSV descriptors to a Ridge metadata model for 48 specimens. Each point is the change in that specimen’s mean absolute terminal error across two held-out evaluations, using the same specimen partitions for both configurations. Negative values favour metadata+HSV; positive values favour metadata alone. There are 19 improvements and 29 worsenings. The dashed line is the specimen-weighted mean, approximately $+0.0002$ kN. Specimens are sorted for readability. This descriptive comparison retains the original model-selection limitations.

6.3 Diagnostics of the metadata baseline

The pooled and paired comparisons establish a relative result: the tested image features do not reliably improve the metadata anchor. They do not establish that the anchor is uniformly accurate. The following diagnostics examine its absolute errors, fitted coefficients and sensitivity to the number of training specimens, using only saved outputs from the same pooled experiment.

6.3.1 Where terminal predictions err

Figure 6.3 averages each specimen’s two held-out terminal predictions before plotting parity and signed residuals. Both mesh groups contain substantial departures from the identity line. The broad separation between groups is compatible with the design structure examined in Section 6.5; it does not guarantee fine discrimination between specimens within either group. A competitive mean error can therefore coexist with consequential individual errors.

This aggregation has MAE 0.171941 kN, while the primary arithmetic mean of fold MAEs remains 0.172711 kN. The paired HSV comparison instead averages absolute errors before comparing models. Averaging signed predictions can cancel some repeat-to-repeat error, so these summaries answer different questions. The original diagnostic’s error bars describe variation between two overlapping-training-set predictions; they are omitted here because they are not calibrated prediction intervals. Repeat agreement alone does not establish accuracy.

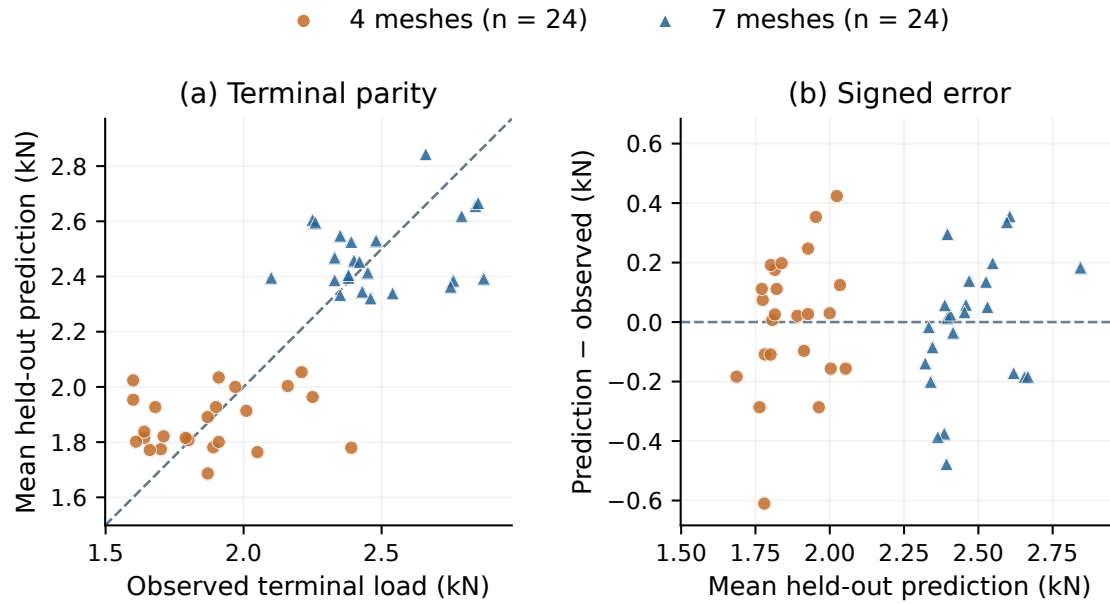


Figure 6.3: Metadata-only Ridge diagnostics for all 48 terminal specimens in the pooled repeated five-fold experiment. Each point averages the specimen's two held-out predictions; orange circles denote four meshes and blue triangles seven meshes, with 24 specimens in each group. Left: predicted versus observed load, with an identity line and equal axes covering all measured values from 1.60 to 2.87 kN. Right: prediction minus observation versus prediction, with zero-error reference. These are post-selection descriptive diagnostics, not independent validation or early-horizon forecasts. No calibrated prediction intervals are available.

6.3.2 What the saved coefficient summary can establish

Figure 6.4 describes a separate full-data refit of the selected Ridge configuration. All encoded columns are standardized; the saved summary sums absolute coefficients over the encoded columns belonging to each original field. Treatment group, series and treatment protocol have the largest aggregated magnitudes. The two observation-time fields are near zero. This makes the fitted representation concrete, but does not identify independent engineering drivers.

The same experimental information can appear in multiple correlated fields, and categorical sums also depend on the number of encoded levels. Consequently, similar coefficients for related fields do not demonstrate separate contributions. Near-zero time coefficients do not show that exposure time lacks a physical effect: every training photograph carries the same terminal target within its specimen. The cover field also retains the information-availability limitation described in the methods chapter. These restrictions connect the coefficient picture to the confounding analysis rather than explaining away that limitation.

6.3.3 Sensitivity to the number of training specimens

The saved learning-curve experiment samples whole training specimens at five fractions within each of the ten grouped splits. Figure 6.5 retains all individual terminal MAEs and their descriptive mean and sample SD. The mean held-out MAE falls from 0.332 kN with eight training specimens to 0.173 kN with 38 or 39, while training-terminal MAE rises from 0.037 to 0.119 kN. At the largest size, a train/test gap remains, and small-subset results include substantial fold variation.

This is evidence of sensitivity within the saved sample, not a forecast of the benefit of collecting more specimens. Sample composition changes with each subset, the same held-out sets are reused across fractions, and no new campaign is introduced. Thus the curve is consistent with limited training

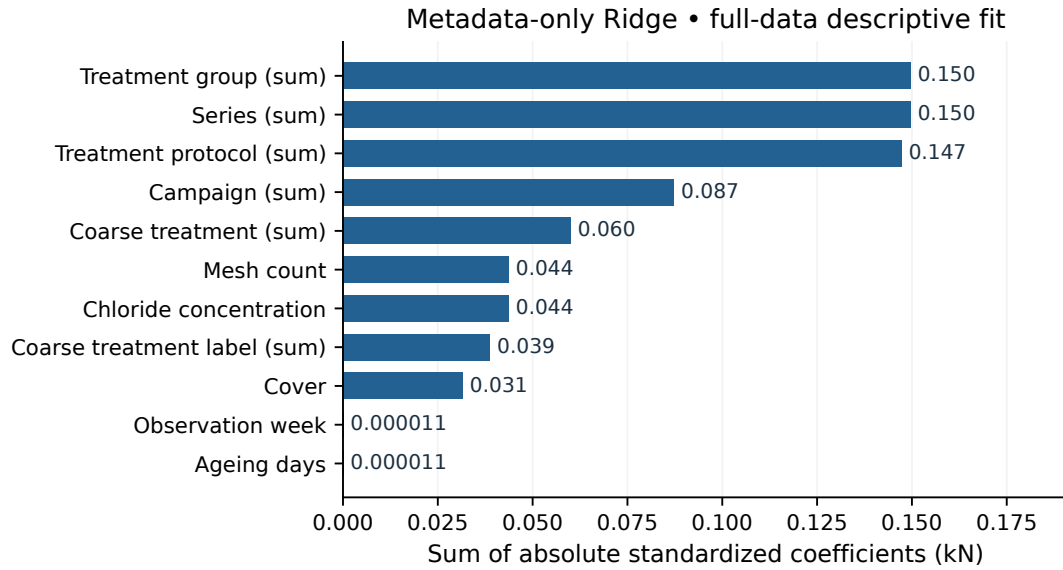


Figure 6.4: Absolute standardized coefficient magnitudes from the saved full-data metadata-only Ridge fit on the pooled photographic histories of 48 specimens. All 11 original metadata fields are retained. Labels marked “sum” aggregate absolute coefficients across standardized one-hot columns; the remaining fields each have one numeric coefficient. Near-zero coefficients are labelled numerically. Magnitudes are in kN per standardized encoded coordinate, summed within field. Category count, correlated inputs and regularization affect this ranking. It is a descriptive full-fit summary, not held-out importance, an independent feature effect or a causal decomposition.

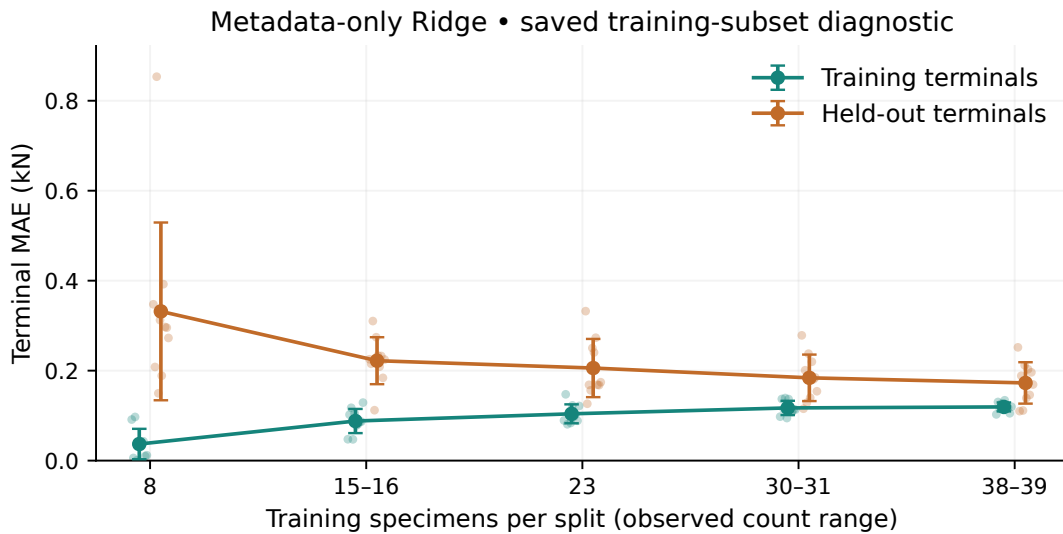


Figure 6.5: Saved metadata-only Ridge learning-curve diagnostic: 50 subset/split runs, comprising five training fractions in each of ten pooled grouped folds. Models were originally fitted on all retained image rows of the sampled training specimens with repeated terminal targets; both training and test errors shown here are scored at terminal observations. Faint points show all individual fold values with horizontal offsets for visibility; connected points and bars show arithmetic means and sample SD. Tick labels give actual training-specimen count ranges. Test folds contain nine or ten specimens. Repeated folds are dependent, subsets need not be nested, and the bars are not confidence intervals. The redraw performs no training.

information, but it does not isolate sample size as the cause of error or resolve the transfer limitation. Chapter 8 brings these diagnostics back to the interpretation of metadata dominance.

6.4 The limited post-onset result

The post-onset sensitivity analysis retains 43 terminal specimens after the first total-rust observation of at least 0.5%. Its metadata+RGB+HSV CatBoost configuration has mean MAE 0.163 kN and $R^2 = 0.660$, compared with metadata-only Ridge MAE 0.173 kN and $R^2 = 0.633$. Mean Spearman is 0.791 for the combined model and 0.803 for metadata alone. Thus the observed improvement is limited to some metrics, rather than a uniform increase in predictive quality.

The selected model family changes as well as the input features, and the subset is smaller than the primary pooled basis. This is an exploratory sensitivity result. It motivates a future matched-model, predeclared study of image availability after onset; it does not overturn the pooled finding or establish an engineering-relevant gain.

6.5 Design stratification and campaign exclusion

Figure 6.6 combines the campaign design with specimen-level terminal associations. Total surface rust and load have pooled Spearman correlation 0.462 for 48 specimens. Within the 24-specimen mesh groups, the corresponding correlations are -0.046 for four meshes and -0.401 for seven meshes. The first is approximately null; the second is negative. This demonstrates the sensitivity of the pooled association to the group structure without decomposing its causal components.

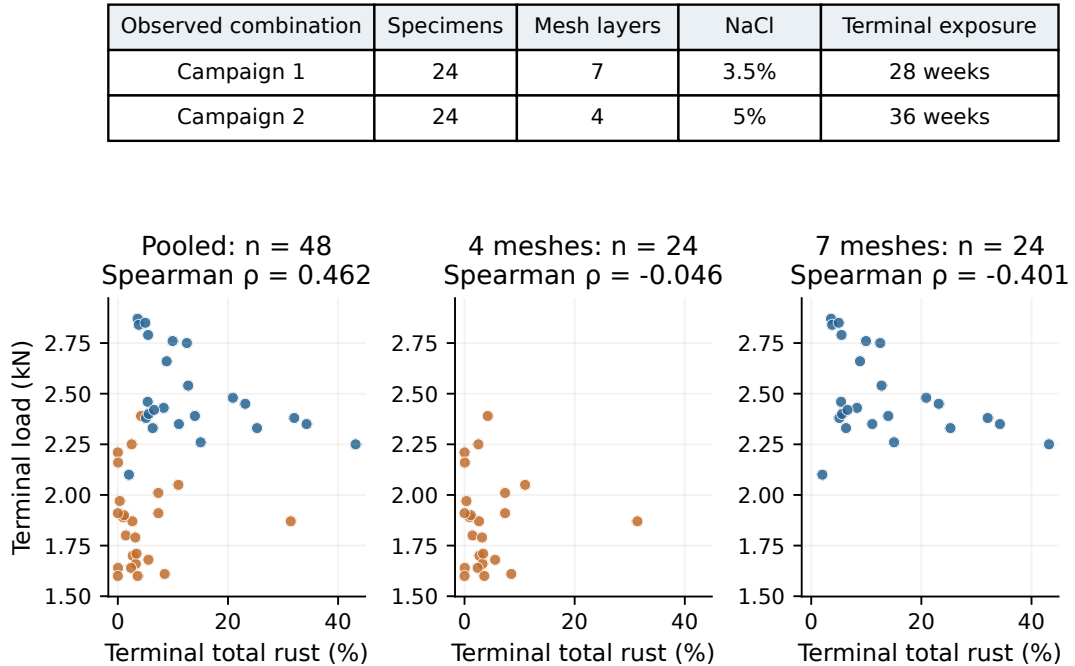


Figure 6.6: Campaign design and terminal surface-rust/load associations. The table shows the two observed combinations of mesh, chloride, and terminal exposure. Scatter plots use one terminal observation per specimen: pooled $n = 48$, four-mesh $n = 24$, seven-mesh $n = 24$. Orange points are four-mesh specimens; blue points are seven-mesh specimens. Correlations are independently recomputed from the saved specimen summary. The pooled positive association does not establish an independent effect of rust on load.

All selected feature-family configurations have negative mean test R^2 within either mesh family. In

the four-mesh group, values range from -3.557 to -0.806 ; in the seven-mesh group they range from -4.654 to -0.298 . Absolute MAE can still look moderate because the target range is narrower. For example, the selected seven-mesh HSV model has MAE 0.137 kN and Spearman 0.460 , but mean $R^2 = -0.386$. Such a result does not justify the claim that every subgroup model contains no ranking information; it does limit a claim of reliably accurate capacity prediction. Full subgroup results are retained in Appendix A.

Under campaign exclusion, the best metadata-only family uses GradientBoosting, with mean MAE 0.465 kN and mean $R^2 = -4.639$ over the two directions. This differs from the pooled Ridge selection, so dividing these two scores would not be a fixed-model robustness ratio. The result shows poor performance even after choosing the best metadata-only model within this stress-test evaluation.

6.6 Residual correction and structural conclusion

A specimen-summary refinement tests whether corrosion summaries can correct metadata-prediction residuals within mesh families. Its selected pipeline applies no residual correction: stage-one Ridge alone gives mean MAE approximately 0.173 kN and mean Spearman 0.749 . The attempted correction models do not improve that baseline in the saved comparison. This is a separate corroborating analysis within the same dataset, not external validation.

The structural evidence converges on a limited capability: pooled prediction can exploit specimen and exposure metadata, while incremental image value and transfer are not established reliably. Sparse terminal supervision and campaign confounding are consistent with the weakness, but the experiments do not isolate their relative contribution from representation limitations or unmeasured variation. The following chapter examines what happens when these structural estimates are propagated into time-dependent screening.

Chapter 7

Degradation and Threshold-Status Analysis

This chapter answers RQ5 by tracing the source of each quantity in the downstream pipeline. A computationally complete curve-and-threshold workflow is not itself validation of the physical trajectory or of remaining useful life.

7.1 Observed, estimated, and derived quantities

The photographic time series supplies repeated visible-corrosion observations. The structural model supplies estimated wire-area loss or load at times where no structural test was performed. A curve fitted to such estimates is one further modelling layer. A crossing time derived from that curve and a chosen threshold is another.

For an estimated structural signal $\hat{a}_i(t)$, a fitted curve $h_i(t)$, and a threshold τ , a candidate crossing is

$$t_i^*(\tau) = \inf\{t : h_i(t) \geq \tau\}. \quad (7.1)$$

A forward interval can be reported only when t_i^* is later than the last observation and falls within the chosen projection horizon. Crossing before or during observation is a different status. Failure to cross within a finite projection is a model/horizon outcome, not evidence that the real specimen survives indefinitely.

The earlier implementation fitted curves to both measured surface series and estimated structural series. The refined implementation focuses threshold screening on estimated wire-area-loss fraction. These versions differ in signal, thresholds, and horizon; their crossing counts should not be interpreted as a controlled improvement in prognostic accuracy.

7.2 Refined saved output

The refined curve-selection output contains 48 fits: 27 linear and 21 monotone-isotonic. Smoothness or monotonicity is a modelling constraint, not evidence that the inferred path reproduces the specimen's actual hidden degradation. The structural models were refitted on available structural data before generating these trajectories; the saved degradation inputs are not fully out-of-fold structural predictions.

Figure 7.1 reports the status of all 144 specimen–threshold combinations for wire-area-loss thresholds 0.20, 0.30, and 0.40 and a 365-day projection horizon. There are zero future crossings at every threshold. Across threshold combinations, 45 are already crossed at baseline, two cross during observation, and 97 do not cross within the horizon. These counts are combinations, not counts of distinct specimens across all thresholds.

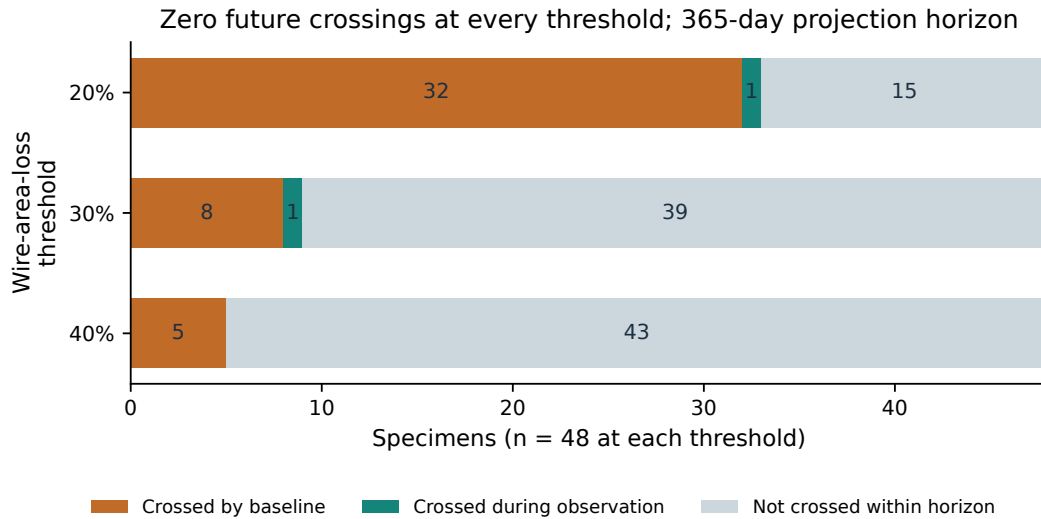


Figure 7.1: Refined threshold-status distribution from estimated wire-area-loss trajectories. Each horizontal bar contains 48 specimens at one exploratory threshold. Crossings already present at baseline and during observation are retained, while no crossing is forecast after the last observation within the 365-day horizon. Grey cases do not cross within that model horizon. These are downstream model statuses, not observed structural failures or calibrated survival outcomes.

7.3 Why this is not validated prognosis

Neither the inferred structural paths nor the threshold meanings have independent longitudinal validation. Refitting on all terminal measurements creates an optimism risk if subsequent trajectories are read as out-of-sample predictions. Replacing those inputs with held-out estimates would address part of that information-flow problem, but would not validate an intermediate structural path or establish that the selected thresholds define service-life failure.

The baseline-crossed cases are particularly diagnostic. Some estimated wire-loss levels are high before substantial visible corrosion is recorded, which motivates a physical-plausibility review. This is an unresolved diagnostic concern, not an experimentally established explanation of each baseline crossing. The correct response is to inspect the model and measurement chain, not to recast baseline crossing as zero remaining useful life.

7.4 Contribution of the downstream analysis

The useful output is an explicit account of signal provenance, fitting assumptions, and crossing status. It makes the dependence on structural estimates visible and demonstrates where validation data are missing. It supports methodological development and diagnostic screening of the computational chain. It does not supply a validated prognostic score or an operational maintenance interval. The discussion now integrates this boundary with the surface and capacity findings.

Chapter 8

Discussion

The evidence is most useful when interpreted as a hierarchy of questions with different supervision requirements. This chapter separates direct observations from plausible explanations and the research decisions they motivate.

8.1 What the supervision asymmetry explains, and what it does not

Visible labels accompany repeated photographs, so a surface model can be evaluated throughout the observed visual range. Structural endpoints provide only one measured response per specimen and are embedded in a confounded experimental design. This difference is consistent with stronger visible-condition estimation and weaker structural transfer.

It is tempting to conclude that sparse labels alone explain all structural weakness. The available experiments do not support that decomposition. The saved learning curve varies the number of training specimens, but subset composition changes with it. It does not separate sample size from design coverage or measure all potentially relevant physical factors. Surface appearance may omit information about local metal loss, bond, geometry, or cracking; the selected feature families may also fail to capture available information. The evidence identifies these as unresolved possibilities rather than estimated causes.

The same discipline applies to label adjacency. Near reconstruction explains why a particular total-rust benchmark needs modest interpretation. It does not erase the practical utility of automated surface measurement, and it does not make every peak-rust result tautological. Target-specific evidence should govern the scope of each concern.

8.2 Two compatible readings of metadata dominance

Metadata can carry physically meaningful design information. A mesh configuration and exposure history may legitimately inform a capacity prediction. At the same time, metadata can identify the campaign mixture because those factors co-vary. The pooled results cannot decide how much predictive value belongs to each reading.

Mesh-stratified and campaign-out evaluations narrow what can be claimed. Within a mesh family, a model has less between-group structure to exploit. Under campaign exclusion, it must operate on a combination absent from training. The degraded results limit a claim of transferable specimen-level discrimination. They do not establish that metadata are invalid inputs or that visible corrosion has no physical relationship with capacity.

The baseline diagnostics make this distinction visible. Figure 6.3 shows individual terminal errors behind the mean score. Figure 6.4 shows how the full fit allocates coefficient magnitudes among correlated metadata fields without identifying independent effects. Figure 6.5 shows sensitivity to the number and composition of training specimens within the same mixture. None supplies external validation, calibrated uncertainty or a decomposition of the causes of structural error.

The incremental-value finding should likewise be read narrowly. Image descriptors are tested against a strong metadata baseline for terminal load. The absence of a reliable pooled increment in the saved comparisons is an empirical limit under those inputs, partitions, and selection rules. It is not

proof of a universal zero effect; the small post-onset result is one reason to retain that distinction.

8.3 Why terminal-load refocusing is scientifically useful

Terminal load is one of two directly measured structural endpoints. Selecting it as the primary target shortens the validation chain: predictions can be compared with an actual terminal measurement, and image value can be tested against metadata. This permits a sharper question than attempting to validate a succession of inferred hidden states, fitted curves, and crossing times without intermediate ground truth.

The refocus does not turn terminal-image prediction into prognosis. Its primary score is evaluated at the terminal observation, and its training label remains eventual load even for earlier images. A future early-warning study needs a different evaluation design with explicitly restricted observation times. Recognizing this boundary is part of the value of the focused analysis.

8.4 Scientific value of the negative findings

The negative results identify different constraints. The pooled ablation limits the tested image increment. The within-mesh results show how an apparently reasonable MAE can coexist with poor error relative to a narrow target distribution. LOCO exposes extrapolation beyond the observed design combinations. The residual-refinement failure suggests that the tested correction does not resolve the remaining specimen-level gap. The threshold statuses show that a working downstream pipeline need not produce forward crossing estimates.

These findings are mutually informative, but they are not all independent replications. Most share the same specimens; the full image/metadata ablation is centralized in one modelling generation. The report therefore treats agreement as convergence of diagnostics within a dataset, not evidence of broad external validity.

8.5 A practical interpretation for research use

The work can support surface quantification research, comparative evaluation, and decisions about what data should be collected next. It does not support direct capacity certification from a photograph. The distinction follows from the demonstrated evaluation scope, not from a generic objection to machine learning.

The natural next step is to align each desired claim with the measurement and validation needed to test it. Chapter 9 makes the outstanding limitations explicit, and Chapter 10 converts them into a concrete research agenda.

Chapter 9

Limitations and Threats to Validity

The limitations are part of the central result because they determine which uses the evidence can support. This chapter groups them by the part of the inference chain they constrain.

9.1 Experimental design and independent sample size

There are only 48 independent specimens and two campaigns. Repeated photographs improve the visual record but do not increase the structural sample size. Mesh, chloride, and terminal duration are aligned with campaign, preventing identification of their separate effects in the pooled data. Treatment variation remains, but small and uneven groups further restrict transfer evaluation.

The specimens and laboratory photography do not represent a sampled population of in-service structures. The study does not estimate performance under changes in camera, illumination, cementitious substrate, construction history, or field exposure. An additional random split of the same observations would not establish those forms of validity.

9.2 Labels and measurement meaning

Visible-rust labels and image descriptors may be closely aligned by construction or measurement procedure. The observed near-identity for total rust limits claims of independent predictive discovery. Exact equivalence of the historical image-processing recipe was not established. The current reconstruction also cannot recover every undocumented predecessor preprocessing choice.

Wire-area loss concerns a selected cross-sectional measurement, not a fully observed internal-damage field. Ultimate load is a terminal structural outcome. Both are measured once per specimen, and neither supplies time-varying structural ground truth. The fraction-versus-percent convention must remain explicit. Category schemas differ between historical builds, so apparent classification improvements across them are not interpretable as controlled gains.

9.3 Evaluation, selection, and statistical uncertainty

Specimen-disjoint splits prevent a direct overlap of specimen histories, but they do not remove campaign confounding, target-adjacent features, or selection bias. Earlier image-fallback and imputation procedures are historical implementation limitations rather than evidence that all preprocessing was fitted independently in every old benchmark. The focused capacity implementation fits its preprocessing within training folds.

Per-family model selection and reporting use the same evaluation folds. No untouched external confirmation set validates the selected structural pipeline. Fold standard deviations describe variation over dependent splits; they are not sampling confidence intervals. Small within-mesh test sets make R^2 and rank correlation unstable, and a mean over folds can hide asymmetric campaign or treatment failures.

A negative mean R^2 is evidence of poor performance under that score and partitioning, not a theorem that every prediction is useless. Conversely, moderate pooled MAE does not establish calibration, useful ranking within a subgroup, or adequate performance for a structural decision.

9.4 Prediction time and structural modelling

The primary load experiment scores terminal images, so it does not quantify a fixed lead time before testing. Training with repeated terminal labels is a legitimate formulation only when the target is stated as eventual terminal load. It is not supervision for current capacity at every photographed time.

The row-level training objective gives more influence to specimens with more retained images. A separate specimen-summary experiment partly addresses the representation issue, but is not a general proof that this weighting is harmless. Metadata definitions include design, treatment, and exposure information; the availability and measurement timing of each field in a future deployment need independent assessment. In particular, cover is recorded as a failure-surface quantity; the saved feature definition does not establish that its value would have been known before destructive testing. No early-use claim should assume that availability without verification.

9.5 Degradation and threshold screening

Structural trajectories are model-generated and partly informed by refits to terminal data. Their fitted smoothness is not validated physical degradation. Threshold choices are exploratory, and no lifetime-event labels validate threshold crossing as failure. Zero future crossings must not be shortened to zero crossings: many combinations are already crossed at baseline or during observation. Model-horizon non-crossing is not an observed right-censored survival outcome.

9.6 Four-class classification branch

The branch is prepared but untrained. Its severe class imbalance and small validation/test specimen counts remain unresolved performance issues. Augmented images share information with their originals and do not create independent specimens. The manifests protect specimen separation and keep validation/test originals unaugmented, but these properties alone do not establish classifier accuracy or robustness.

9.7 Software and reproducibility

Two current issues constrain model regeneration: unquoted YAML `NO` becomes boolean false for the no-treatment group, and numeral/word substitutions involving `main_first` remain in source and configuration files. The cited saved modelling outputs predate the affected source modification timestamps, supporting the historical-output interpretation. Modification times do not establish the exact historical executable, root cause, or freedom from all earlier errors.

The present rebuild reproduces aggregation and figures from saved data and checks immutable-report hashes. It does not claim push-button reproduction of the historical training. Historical hardware, exact package environments, and full compute budgets are not consistently archived. These constraints are documented with concrete source paths in Appendix A.

Taken together, the limitations support a focused scientific conclusion rather than an operational predictor. The final chapter states what is completed, what remains prepared, and what additional evidence each stronger claim would require.

Chapter 10

Current Status, Next Steps, and Conclusions

The completed evidence supports a research synthesis and a bounded empirical claim. This chapter distinguishes that completed work from the prepared classification branch and from future experiments that require additional validation or data.

10.1 Current research status

The surface and terminal-structure modelling experiments have saved results. The full image-versus-metadata comparison for terminal load has been completed and rechecked from its prediction files. The degradation pipeline produces model-derived trajectories and threshold statuses. The current four-class classifier has not been trained or evaluated.

Table 10.1: Prepared four-class total-rust manifests. Counts come from the actual retained manifests, not the larger global augmentation inventory. All 48 specimens are assigned to disjoint splits; validation and test contain originals only. These are data-preparation counts and imply no classification performance.

Split	Specimens	Originals	Augmented	Total
train	38	641	3205	3846
val	5	75	0	75
test	5	75	0	75

The training set contains 3,846 images from 38 specimens, comprising 641 originals and 3,205 augmented copies. Validation and test each contain 75 original images from five specimens. These numbers differ from the complete offline augmentation inventory because transformed copies from held-out specimens are excluded from training. The final retained manifests contain 3,996 images in total.

10.2 Next steps that can use the existing data

1. **Restore reproducible model inputs.** Quote categorical values explicitly, repair the identified substitution errors through targeted tests, and archive exact feature lists and environment versions. Compare repaired-run outputs with saved artifacts before treating any changed metric as a scientific improvement.
2. **Run the four-class baseline.** Keep specimen-disjoint manifests fixed and use training-only augmentation. Report macro-F1, class-specific recall, confusion matrices, and a simple class-frequency baseline. Test labels should remain untouched during model selection. These metrics are a proposed evaluation plan, not obtained results.
3. **Test an actual early prediction question.** Choose observation cutoffs before training and use only information available by each cutoff. Predict terminal load with specimen grouping and a metadata anchor. This would evaluate lead-time usefulness directly, which terminal-image scoring does not.

4. **Separate feature effects from model choice.** Compare adding images within a fixed model family and evaluate the entire selection process with a confirmation strategy appropriate to the small sample. Report paired specimen-level differences alongside fold summaries without treating repeated predictions as independent outcomes.
5. **Improve downstream diagnostics.** Generate structural estimates without fitting on the evaluated specimen, expose raw versus smoothed trajectories, and separately review physical threshold justification. These changes can improve methodological integrity but cannot manufacture missing structural trajectories or lifetime outcomes.

10.3 Data needed for stronger structural and prognostic claims

Additional campaigns should vary mesh, chloride, and exposure duration across overlapping combinations rather than preserve their current alignment. Such a design would make transport and subgroup performance more interpretable. A model trained on a subset of combinations should be evaluated on explicitly described held-out combinations.

Repeated structural information requires an experimental strategy compatible with destructive tests, such as scheduled testing of comparable specimens or independent nondestructive structural measurements validated against terminal outcomes. Such measurements would still need a defensible link between observed and inferred condition. Prognostic validation further requires a defined service-life endpoint and observed event or censoring information. Neither requirement is met by appending more derived threshold columns to the existing table.

10.4 Answers to the research questions

RQ1: Visible corrosion is learnable across distinct representations, with strong label adjacency in the refined total-rust benchmark. The result supports automated surface estimation within the observed label system.

RQ2: Terminal structural inference is less stable. Final robustness-weighted models use metadata alone, and stricter or stratified evaluation limits claims of transferable image-based structural prediction.

RQ3: The focused pooled experiment does not establish reliable incremental image value for terminal load. The post-onset sensitivity has a limited improvement in some metrics and does not overturn that conclusion.

RQ4: Evaluation regime changes the claim. Campaign exclusion is a severe extrapolation stress test, and visible-treatment transfer is phase-dependent. Grouped performance alone does not establish external transfer.

RQ5: The degradation and threshold pipeline is methodological and diagnostic. It uses estimated structural paths and yields zero future crossings in its refined configuration; it does not validate remaining useful life.

10.5 Closing conclusion

The principal result is a calibrated boundary: photographic surface assessment is supported more directly than structural inference, and structural inference is supported more directly than lifetime prediction. This hierarchy is an interpretation of the present evidence, not a universal ordering of possible methods. By tying claims to target definitions, independent specimen counts, metadata controls, and evaluation regimes, the activity provides a defensible basis for the next experiment. The appendices preserve the implementation detail needed to inspect and reproduce that account.

Appendix A

Reproducibility and Supplementary Results

This appendix records the saved evidence and reproduction boundary underlying the scientific narrative. Paths are relative to the repository root; they are implementation details rather than additional scientific claims.

A.1 Evidence source map

Table A.1: Saved repository sources for the report. Paths are relative to the repository root.

Claim or output	Source
Dataset and terminal labels	main_4/outputs/data/master_table.csv and main_4/outputs/audit/verified_facts.json
Campaign mapping	main_4/configs/specimen_mapping.yaml
Classical visible benchmark	main/reports/current_corrosion_metrics.csv
Deep representation	main_2/reports/model_comparison.csv; main_2/reports/predictions_best_model.csv; main_2/artifacts/run_info.json
Interpretable benchmarks	main_3/outputs/corrosion_regression_metrics.csv; main_3/outputs/damage_regression_metrics.csv
Label adjacency	main_4/outputs/features/image_features.csv, joined to the aligned master table by sample name
Fixed-model robustness	main_4/outputs/diagnostics/tables/benchmark_best_model_r obustness.csv and earlier per-fold CSVs
Final structural selection	main_4/outputs/models/hidden_damage/best_models.csv
Capacity ablation	main_4/outputs/ultimate_load_refocus/pooled_all_weeks/gr ouped_cv/feature_set_comparison.csv; each selected configuration's terminal_fold_predictions.csv and fold_metrics.csv
Stratified and sensitivity analyses	main_4/outputs/ultimate_load_refocus/mesh_stratified/mes h_feature_set_comparison.csv; main_4/outputs/ultimate_load_refocus/pooled_post_onset/g rouped_cv/selected_best_model_per_feature_set.csv
Terminal associations	main_4/outputs/ultimate_load_refocus/data/specimen_summa ry_table.csv
Residual refinement	main_4/outputs/ultimate_load_refocus/residual_refinement /model_comparison.csv and best_model.csv
Threshold statuses	main_4/outputs/models/proxy_rul/proxy_rul_estimates.csv; proxy_rul_summary.json in the same directory; main_4/configs/thresholds.yaml
Four-class preparation	Data/splits/train_manifest.csv, val_manifest.csv, test_manifest.csv; augmentation/make_splits.py
New verification and figure records	report_v2/VERIFICATION_LOG.md; report_v2/evidence/claims.json; report_v2/figures/PROVENANCE.md

A.2 Feature and computation contract

The focused capacity metadata list contains campaign, series, treatment group, treatment protocol, coarse treatment, coarse treatment label, mesh count, chloride concentration, cover, observation week, and ageing days. These are 11 input columns before encoding, not necessarily 11 dimensions after one-hot encoding. RGB and HSV families contain 41 and 36 input columns, respectively. The combined metadata+RGB+HSV input contains 88 columns before encoding. These counts are taken from the saved feature-family comparison, not inferred from the number of displayed importances.

The image-only names in that experiment refer to derived RGB/HSV families with the manually assigned visible targets excluded. The earlier mixed structural representation does include recorded visible labels. These input contracts prevent the phrase “image-based structural model” from erasing materially different information sources.

Saved configurations specify a random seed of 42 for the focused experiments, two repetitions of five folds, and mesh stratification in the pooled case. The original model families include tree ensembles and Ridge. Some numeric configuration fields are currently corrupted, so this appendix does not silently substitute guessed historical hyperparameter values. The saved results and exact current files are hashed in the verification tables. Historical training duration and complete environment metadata are not consistently available; the report does not invent them.

A.3 Full sensitivity tables

Table A.2: Selected post-onset configurations for 43 terminal specimens over ten grouped folds. MAE and sample SD are in kN; other columns are mean fold metrics. Input/model selection is exploratory and shares the evaluation folds.

Features	Model	MAE	SD	R^2	ρ
metadata+RGB+HSV	CatBoost	0.163	0.022	0.660	0.791
metadata-only	Ridge	0.173	0.046	0.633	0.803
metadata+HSV	Ridge	0.182	0.035	0.572	0.817
HSV-only	CatBoost	0.197	0.046	0.516	0.738
metadata+RGB	Ridge	0.209	0.034	0.453	0.770
RGB+HSV	CatBoost	0.210	0.034	0.505	0.772
RGB-only	CatBoost	0.234	0.053	0.405	0.660

Table A.3: All selected within-mesh feature families, with 24 terminal specimens per mesh group and ten grouped folds per analysis. MAE is in kN. Every row has negative mean test R^2 ; Spearman varies and is not uniformly null. Model choices differ by feature family.

Subset	Features	Model	MAE	R^2	ρ
mesh_4	metadata+RGB+HSV	CatBoost	0.194	-0.998	0.081
mesh_4	RGB-only	CatBoost	0.203	-0.906	0.097
mesh_4	metadata-only	Ridge	0.208	-1.476	0.301
mesh_4	HSV-only	CatBoost	0.210	-0.806	0.057
mesh_4	RGB+HSV	CatBoost	0.214	-1.050	0.187
mesh_4	metadata+HSV	Ridge	0.228	-1.931	-0.025
mesh_4	metadata+RGB	Ridge	0.261	-3.557	0.085
mesh_7	HSV-only	CatBoost	0.137	-0.386	0.460
mesh_7	RGB+HSV	CatBoost	0.138	-0.298	0.330
mesh_7	RGB-only	CatBoost	0.140	-0.407	0.290
mesh_7	metadata+RGB+HSV	CatBoost	0.204	-1.980	0.010
mesh_7	metadata+RGB	Ridge	0.214	-4.654	0.030
mesh_7	metadata-only	Ridge	0.216	-3.093	-0.050
mesh_7	metadata+HSV	Ridge	0.245	-3.639	-0.000

The paired diagnostic added after first-draft review reads the metadata-only Ridge and metadata+HSV Ridge terminal-fold prediction files under the same pooled experiment. It verifies matching specimen/split keys and equal target values, averages absolute errors over each specimen’s two held-out evaluations, and then weights the 48 specimen differences equally. The complete paired data and source hashes are saved in `report_v2/tables/paired_specimen_errors.csv` and `report_v2/evidence/paired_diagnostics.json`. This is a new aggregation of saved predictions, not a model refit.

Table A.4: Threshold status of model-derived wire-area-loss trajectories for 48 specimens per threshold. Values are counts, not measured failures. Thresholds are fractions expressed here as percentages. “Not crossed” means within the configured 365-day projection horizon; the term does not imply a validated survival observation. Future crossing count is zero at all thresholds.

Threshold	Baseline	During	Future	Not crossed
20%	32	1	0	15
30%	8	1	0	39
40%	5	0	0	43

A.4 Current software defects and chronology

PyYAML parses the unquoted value `NO` as boolean false for ten no-treatment specimens. The saved coarse-treatment field contains the string `False` for 168 rows, while the raw treatment field retains `NO`. The report uses intended treatment semantics when describing the experiment, without modifying the historical modelling artifacts.

The substitution pattern involving `main_first` is present in `main`, `main_2`, `main_3`, and `main_4`; a scan of the augmentation source finds no such instance. Examples include a corrupted pandas aggregation name and invalid YAML numeric values. Their root cause is not established here. The cited saved scientific output artifacts checked in the rebuild predate the earliest affected source timestamp. This observation supports separation of saved results from currently broken code but is not proof of an exact executable history.

The mapping file is itself later than some saved outputs and contains affected narrative fields. Its

campaign/design fields used here remain intact and agree with the saved aligned table. The independent augmentation work is later still and is checked separately. Archive exports and the older snapshot are not treated as additional experiments merely because a copy exists.

A.5 V3 figure audit and diagnostic provenance

The audit opened all 30 candidate files, including every page of the two earlier manuscript PDFs. Four source figures were selected and consolidated into the three diagnostics in Section 6.3; 26 candidates were rejected, most often for repeating existing evidence. Decisions and reproduction limitations are in `report_v2/thesis/v3/FIGURE_AUDIT.md`.

The added Ridge inputs are `terminal_fold_predictions.csv`, `terminal_oof_predictions.csv`, `feature_importance_full_fit.csv`, `learning_curve_raw.csv`, `learning_curve_summary.csv` and `fold_metrics.csv`, all under `main_4/outputs/ultimate_load_refocus/pooled_all_weeks/grouped_cv/experiments/metadata_only/Ridge/`. Recomputed aggregates, source hashes and plot bounds are saved in `report_v2/evidence/v3/figure_checks.json` and `new_figure_checks.json` in the same evidence directory. Every terminal specimen, coefficient field and individual learning-curve error is retained. No model was refitted.

A.6 Reproduction commands and boundary

From the repository root, use the project scientific Python environment to verify saved inputs and regenerate only the added diagnostics:

```
python report_v2/scripts/audit_saved_figures.py
python report_v2/scripts/make_v3_figures.py
python report_v2/scripts/build_v3_reports.py
python report_v2/scripts/validate_documents.py
```

The first command replays only plotting functions, never training orchestration. The second extends the shared figure set and provenance ledger. The build command compiles only v3; validation reads all versions and writes the v3 QA record. Existing v1/v2 sources, PDFs and figure binaries remain byte-for-byte unchanged. The older general-purpose rebuild scripts should not be run as part of this preservation workflow because they regenerate earlier artifacts.

Compile this thesis directly from its v3 directory using:

```
latexmk -pdf -interaction=nonstopmode -halt-on-error thesis.tex
```

The article uses the analogous command with `article.tex` from its own v3 directory. BibTeX reads the shared bibliography. The validator rejects unresolved references, citations, rerun warnings and overfull boxes, checks source hashes, and verifies preservation separately from compilation.

These steps reproduce presentation and saved-data aggregation. Reproducing historical training would require resolving existing software defects and reconstructing missing environment details, and remains outside this work. The references support background and experimental attribution; the source map and audit support repository-specific empirical results.

Bibliography